

Assignment 2

Rules:

- You may use $\text{\LaTeX}$, Word, or any other text editor but please hand in a **pdf document** with your solution.
- It is recommended to use **R** for solving the assignment. Alternatively, you may use Python, Matlab, or Ox. No Eviews, Stata, or anything else.
- You must write your own code - built-in packages (such as **vars** in R) may only be used to double-check the results. **Please hand in your code in an Appendix. Make sure it is well-documented.**
- You should **answer** the questions - do not just paste estimation output without explanation (except when nothing else is asked). **All results should be presented in a readable, clear and concise way.**
- You should hand in one pdf document for each group of **4 students**.
- The deadline is Friday, March 13 at 23:59 hours.

In this assignment, you are asked to build multiple time series models for quarterly U.S. data on gross domestic product (GDP), consumer price index (CPI), and Federal funds rate (IR), a monetary policy interest rate that is set by the central bank. All three variables are constructed as quarterly time series, the first observation is for 1954Q3 and the last is 2024Q3. The source of the data is **FRED2**. Download the data set **data_assignment2.csv** from Canvas and load it into your software. (At a final stage, you may want to consider to extend the data set up to 2025).

Pre-assignment work:

- Plot the three time series in separate graphs. Make sure the x-axis shows the correct time indices (year/quarter). Transform the two macro series (GDP and CPI) to quarterly log growth rates. Plot the two transformed time series as well.
- Carry out a simple univariate autoregression for the three variables GDP growth, CPI growth (inflation rate) and IR as y_t separately, and consider the autoregressive $\text{AR}(p)$ model

$$y_t = \text{constant} + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + \varepsilon_t.$$

for different maximum lag orders $p = 0, 1, \dots, 5$. Report the least squares estimates of constant and ϕ_j coefficients for the different p in a table with six columns ($p = 0, 1, \dots, 5$) include standard errors and/or t-statistics. Comment on your findings.

- Redo the AR analyses above but using only pre-COVID data (that is drop last 20 observations from the sample). Do the results change very much? Provide comments.

The pre-assignment code in Python and Ox is on Canvas.

Assignment 2:

Consider the vector autoregressive (VAR) model for this assignment.

- (a) (2 point) Estimate a **bivariate** $VAR(3)$ model for GDP growth and inflation rate. Include a constant term. Report the estimated coefficients. Does this VAR model with estimated coefficients fulfill the stability condition? Justify your answer by reporting the appropriate results.
- (b) (4 points) Compute the Akaike information criteria for the $VAR(p)$, for $p = 1, \dots, 5$. Which model is the best choice to describe the data according to the Akaike information criterion?
Hint: In order to make a reasonable comparison, you should set aside **5** pre-sample values, that you also do not use for estimating the lower order VARs. Hence, the sample size used for estimation is always the same.
- (c) (4 points) Using the best VAR model from part (c), calculate and plot the four response functions of your system after impulses of one unit. Interpret the impulse response functions and comment.
- (d) (2 points) Redo items (c) and (d) but using only pre-COVID data (that is drop last 20 observations from the sample for both variables). Do the results change very much ? Provide comments.
- (e) (4 points) Estimate a three-dimensional $VAR(2)$ model that includes GDP growth, inflation rate and federal funds rate. Include a constant term. Report the estimated coefficients. You can base your results on the full data or/and on the pre-COVID data.
- (f) (4 points) Using the estimated three-dimensional $VAR(2)$, test whether the federal funds rate Granger-causes the two macro series as a group. Interpret the test result.